CYBERSECURITY
INTERNSHIP PROGRAM

Cybersecurity Lab Setup

Kali Linux Virtual Lab with NAT Networking

Project Module WK1-PM1 | Week 1 Project Report

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MODULE

WK1-PM1: Lab Setup

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1. Introduction

This report documents the Week 1 internship project module **WK1-PM1 (Lab Setup)** completed as part of the Networkwalks Cybersecurity Internship. The goal of the module is to build a personal cybersecurity testing lab on a laptop or PC, with Kali Linux as the attacking and analysis machine, on a private virtual network with working internet access.

A home lab is the foundation of practical security work. It gives a safe, isolated environment where tools can be run, configurations can be broken and repaired, and snapshots can be used to roll back mistakes, all without any risk to real systems. This report describes the lab that was built, the network configuration in use, and the evidence collected to confirm that it works.

2. Objectives and Scope

The project brief for WK1-PM1 lists the following requirements. The status of each item, based on the evidence collected for this report, is shown in Section 7.

- Build a cybersecurity testing lab on a laptop or PC using a desktop hypervisor.
- Set up Kali Linux as the attacking (hacker) machine.
- Use a private virtual network in the 10.0.0.0/24 subnet.
- Enable clipboard sharing and file drag-and-drop in the virtual machine settings.
- Share the host machine's Downloads folder with the virtual machine.
- Give the Kali Linux machine full internet access and verify it with connectivity tests.

3. Tools and Environment

The values below are taken from the virtual machine settings screen captured in Figure 4.

Component	Details
Hypervisor	VMware Workstation (Windows host)
Guest machine	Kali Linux (virtual machine named "kali Linux")
Guest role	Attacking / analysis machine
Memory	2 GB
Processors	1
Hard disk	20 GB (SCSI)
Installation media	Kali Linux ISO stored in D:\Downloads, attached as virtual CD/DVD (SATA)
Network adapters	3 adapters: NAT, Custom (VMnet2), Custom (VMnet8 NAT)
Other devices	USB controller, sound card, display (auto detect)

4. Network Architecture and IP Plan

The lab uses a private virtual network in the **10.0.0.0/24** range. The virtual machine reaches the internet through the hypervisor's NAT service (VMnet8), which acts as the gateway and keeps the lab traffic separate from the physical network. Addressing observed on the active interface (eth2) is summarised below.

Parameter	Observed value	Source
Active interface	eth2 (state UP)	ip -4 addr show eth2
IPv4 address	10.0.0.128/24	ip -4 addr show eth2
Broadcast	10.0.0.255	ip -4 addr show eth2
Address assignment	Dynamic (DHCP)	Route table: proto dhcp
Default gateway	10.0.0.2	ip route
Local network route	10.0.0.0/24 via eth2	ip route
Hypervisor network	VMnet8 (NAT)	VM settings, Network Adapter 3

Note on the brief: the module brief specifies VirtualBox with a NATNetwork and a manual address of 10.0.0.2/24 for Kali. This lab was built on VMware Workstation, where 10.0.0.2 is used by the NAT gateway itself, so the Kali machine received 10.0.0.128 by DHCP. The subnet (10.0.0.0/24) matches the brief.

5. Implementation Steps

The lab was assembled in the following stages.

5.1 Hypervisor and virtual machine

- Installed the hypervisor on the Windows host.
- Created a new virtual machine named **kali Linux** with 2 GB of memory, 1 processor and a 20 GB disk.
- Attached the Kali Linux installation ISO from the host's Downloads folder as the virtual CD/DVD drive and completed the installation.

5.2 Network configuration

- Added network adapters to the VM, including a custom adapter connected to **VMnet8 (NAT)**.
- Confirmed that the adapter is set to *Connect at power on* so the network is available at boot.
- Verified the address, subnet and default route from inside Kali using the commands below.

```
ip -4 addr show eth2     # interface address and state
ip route                 # default gateway and local routes
ping -c 4 10.0.0.2       # gateway reachability
ping -c 4 google.com     # internet and DNS check
```

5.3 Connectivity verification

- Sent four ICMP echo requests to the gateway to confirm local network reachability.
- Sent four ICMP echo requests to **google.com**, which tests DNS resolution and internet access in a single step.

5.4 Command reference

Command	Purpose
ip -4 addr show eth2	Show the IPv4 address, mask and state of the interface
ip route	Show the default gateway and local network route
ping -c 4 <host>	Send four echo requests and report loss and round-trip time

6. Verification and Evidence

The screenshots below were captured from the running Kali Linux virtual machine and from its settings window.

6.1 Interface address

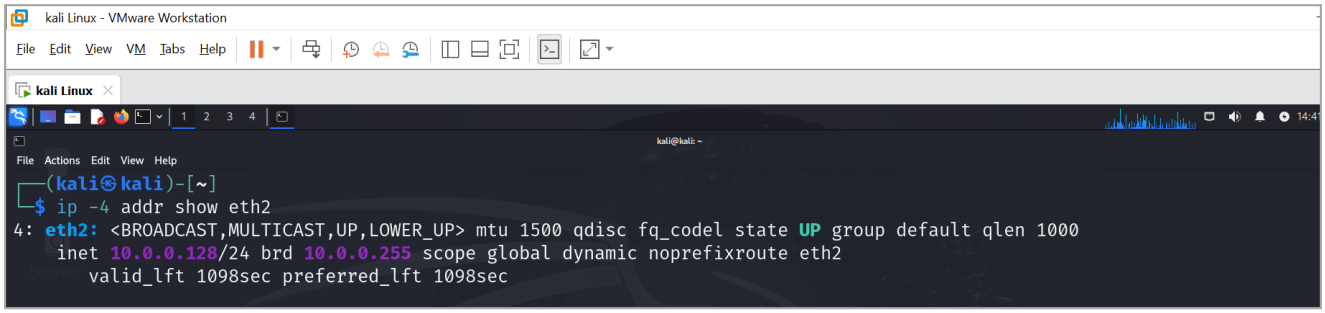


Figure 1. *Output of ip -4 addr show eth2: interface eth2 is UP with address 10.0.0.128/24, assigned dynamically.*

The interface is in the **UP** state with the address **10.0.0.128/24** and broadcast address 10.0.0.255. The *dynamic* flag and the remaining lease time show that the address was obtained from the hypervisor's DHCP service.

6.2 Routing table

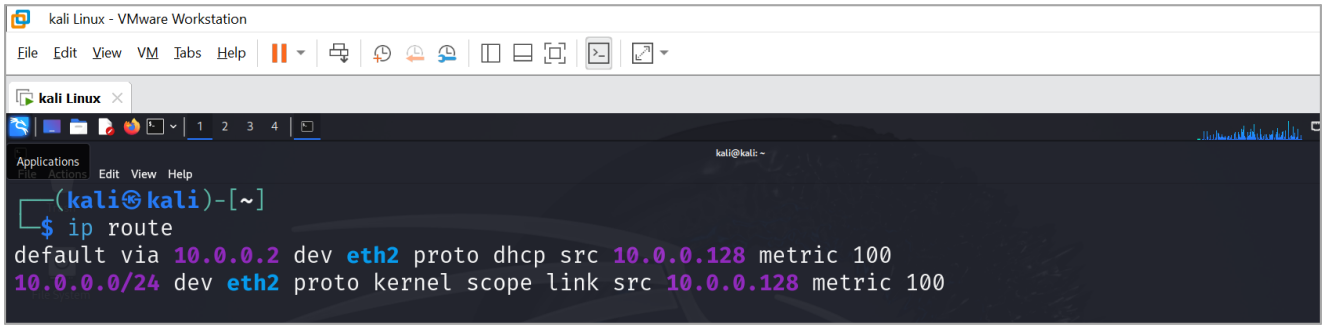


Figure 2. *Output of ip route: default route via 10.0.0.2 and local network 10.0.0.0/24 on eth2.*

The default route points to **10.0.0.2**, the hypervisor's NAT gateway, and the local route confirms that the 10.0.0.0/24 network is attached to eth2. All traffic to the internet is therefore forwarded through this gateway.

6.3 Connectivity tests

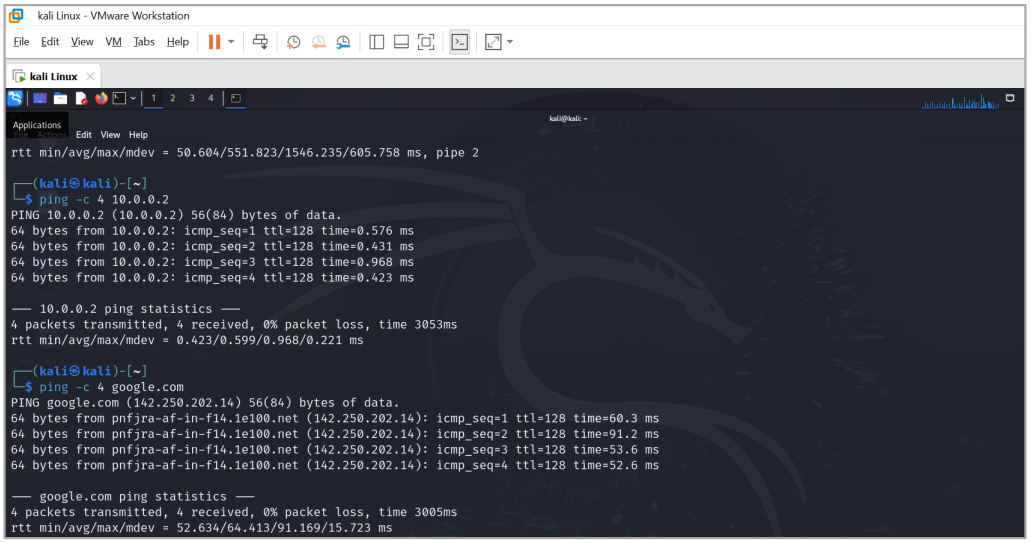


Figure 3. *Ping tests from Kali Linux to the gateway (10.0.0.2) and to google.com (142.250.202.14), 4 packets sent and 4 received, 0% packet loss.*

Test	Command	Packets sent / received	Loss	Avg RTT
Gateway	ping -c 4 10.0.0.2	4 / 4	0%	0.599 ms
Internet (DNS + ICMP)	ping -c 4 google.com	4 / 4	0%	64.413 ms

The gateway replied to all four requests with sub-millisecond latency, showing that the local virtual network is healthy. The ping to **google.com** first resolved the hostname to an IP address, which confirms that DNS is working, and then received all four replies, which confirms outbound internet access through NAT. The higher latency (about 53 to 91 ms) is expected because the traffic leaves the host and crosses the real internet.

6.4 Virtual machine settings

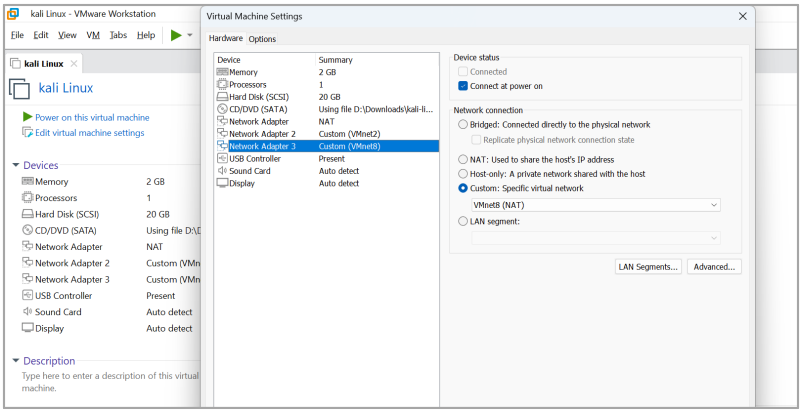


Figure 4. *Virtual Machine Settings: hardware summary and Network Adapter 3 set to Custom (VMnet8 NAT), with Connect at power on enabled.*

7. Test Summary

The table compares the project brief with the results observed in this lab.

#	Requirement / test	Result	Evidence
1	Kali Linux deployed as attacking machine	Done	Fig. 4
2	Network in 10.0.0.0/24 subnet	Done	Fig. 1, 2
3	Gateway reachable	Pass, 0% loss	Fig. 3
4	Full internet access (ping google.com)	Pass, 0% loss	Fig. 3
5	DNS resolution	Pass	Fig. 3
6	Kali address 10.0.0.2/24	Differs: 10.0.0.128/24 via DHCP (10.0.0.2 is the NAT gateway)	Fig. 1
7	VirtualBox with NATNetwork	Differs: VMware Workstation, VMnet8 (NAT)	Fig. 4
8	Clipboard and drag-and-drop enabled	Not shown in this report	-
9	Downloads folder shared from host	Not shown in this report	-

Items 8 and 9 are part of the brief but are not covered by the screenshots collected so far. Evidence for them (the General settings tab for clipboard and drag-and-drop, and the shared folders page) can be added as extra figures when captured.

8. Challenges and Troubleshooting

The following issues are common when building this lab and are recorded here with their fixes.

Issue	Likely cause	Fix
No internet from Kali after boot	Duplicate address detection delay or wrong adapter	Check the adapter is connected to the NAT network; on VirtualBox 7 with Kali 2026.1 or newer, run: sudo nmcli connection modify "Wired connection 1" ipv4.dad-timeout 0, then bring the connection down and up
Gateway ping works but google.com fails	DNS not resolving	Use 8.8.8.8 as the DNS server, or the gateway address if problems continue
Address conflict on the network	Two machines using the same IP	Make sure no other VM uses the same address, and restart the VMs
Interface has no address	Adapter disconnected at boot	Enable Connect at power on and restart the VM

9. Conclusion and Next Steps

A working Kali Linux virtual machine was built on a private 10.0.0.0/24 network behind a NAT gateway. Command-line evidence shows that the interface is up, that a default route exists, that the gateway responds with 0% packet loss, and that the internet is reachable by name with 0% packet loss. This gives a stable base for the practical exercises in later modules.

The lab differs from the brief in two respects: it uses VMware Workstation instead of VirtualBox, and Kali holds a DHCP address (10.0.0.128) instead of the manual 10.0.0.2/24. Both are documented openly in this report.

Recommended next steps

- Take a snapshot of the working Kali configuration as a clean rollback point.
- Capture screenshots of the clipboard, drag-and-drop and shared folder settings to complete the checklist.
- Add further machines (Windows 10, Android-x86) on the same network and run ping tests between all of them.
- Update the system with *sudo apt update && sudo apt full-upgrade* before starting security exercises.

Responsible use: all tools and techniques practised in this lab must only be used on systems you own or have written permission to test.

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